

Problem 25.15

Lauren is being paid a continuous perpetuity payable at a rate of 1000 per year. Calculate the present value of the perpetuity assuming $d^{(12)} = 0.12$.

Solution

We are given the nominal discount rate convertible monthly:

$$d^{(12)} = 0.12.$$

The relationship between the annual effective interest rate i and the nominal discount rate convertible monthly is

$$1 + i = \left(1 - \frac{d^{(12)}}{12}\right)^{-12}.$$

Substitute $d^{(12)} = 0.12$:

$$1 + i = \left(1 - \frac{0.12}{12}\right)^{-12}.$$

Thus,

$$i = \left(1 - \frac{0.12}{12}\right)^{-12} - 1.$$

Therefore,

$$i = 0.128178095.$$

The corresponding force of interest is

$$\delta = \ln(1 + i).$$

So

$$\delta = \ln(1.128178095) = 0.1206040302.$$

A continuous perpetuity payable at rate 1000 per year has present value

$$1000\bar{a}_{\infty|} = 1000\left(\frac{1}{\delta}\right).$$

Thus,

$$PV = \frac{1000}{0.1206040302} = 8291.5968.$$

Therefore, the present value is

$$\boxed{8291.60}.$$